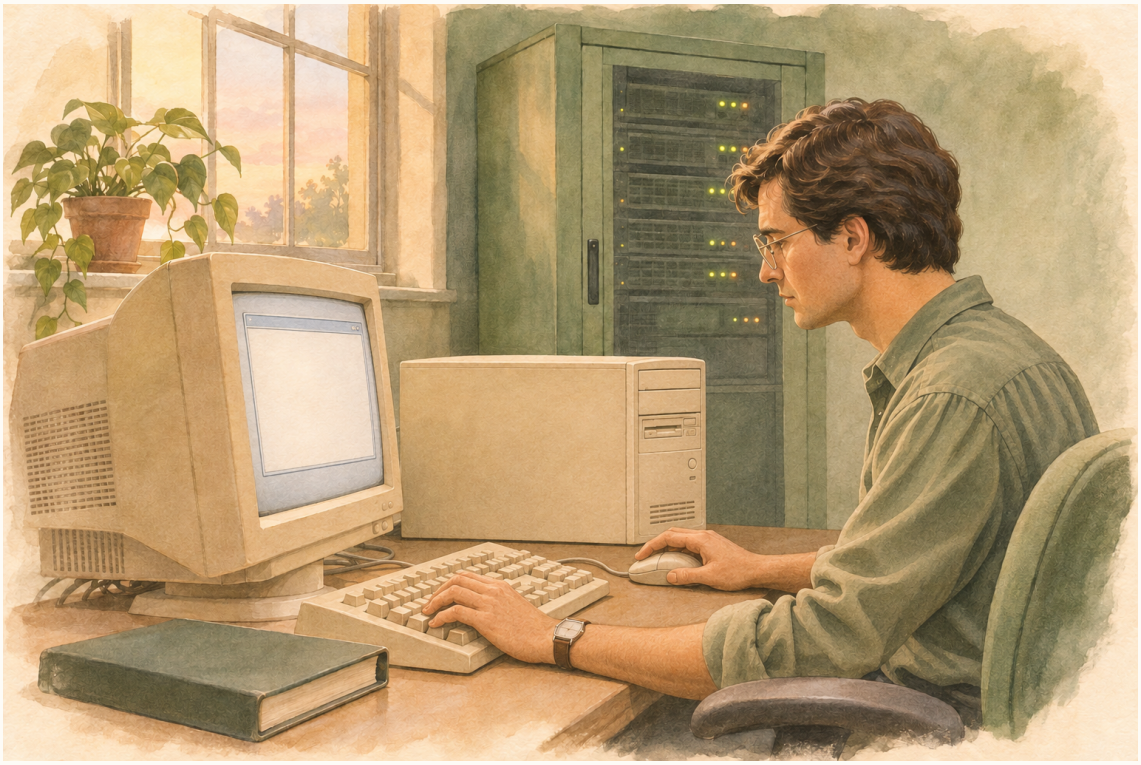
千年虫：被成功抹去的危机

Y2K: The Crisis That Success Erased



危机没有发生，能说明预防多余吗？

英文原文

The Y2K bug, or millennium bug, began as a shortcut. When memory and hardware were costly, programmers stored years as two digits, writing "99" for 1999. That compact code hid an ambiguity: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

Governments and firms braced for chaos. In February 1998 President Clinton created a council to coordinate America's effort. Consultants audited legacy systems and patched faulty date logic. In November 1999 the Commerce Department estimated American governments and enterprises would spend about $100bn; projections of the global bill ran to hundreds of billions.

Then midnight struck on 31 December 1999, and little went wrong. Glitches that did crop up were sporadic and mostly minor. Radiation-monitoring equipment faltered at Japan's Shika nuclear plant without any threat to safety, and a Pentagon ground system that processes spy-satellite data failed. In February 2000 a Senate committee concluded that the expenditure had been justified.

Many saw it differently: the threat had been exaggerated, even a hoax. Yet calm prevailed largely because so much had been fixed. This is the preparedness paradox: when precautions work, the danger they ward off never appears, so it looks imaginary. Tedious repairs earned mockery rather than gratitude.

Critics noted that Italy and South Korea spent far less yet fared no worse. The comparison misleads: those low figures often omitted what firms spent, and much of the remedial work replaced obsolete systems overdue for upgrading. Nobody can say which dollars were essential and which excessive. Prevention that works may leave nothing but an invoice, to be weighed against a catastrophe that might have been.

中文译文

千年虫漏洞最初源于一种省事的做法。当内存和硬件还很昂贵时，程序员只用两位数字储存年份，例如以“99”代表1999年。这种精简代码暗藏歧义：计算机可能把“00”理解成1900年，导致利息、年龄或截止日期计算出错。

各国政府和企业严阵以待，准备应对可能出现的混乱。1998年2月，克林顿总统成立委员会，协调美国的应对工作。顾问们审查旧系统，修补有问题的日期处理逻辑。1999年11月，美国商务部估计，美国各级政府和企业将为此花费约1000亿美元；对全球总支出的预测则高达数千亿美元。

1999年12月31日午夜来临，结果几乎没有出大问题。偶尔出现的小故障也大多轻微。日本志贺核电站的辐射监测设备失灵，但没有威胁安全；美国国防部一套处理间谍卫星数据的地面系统也出了故障。2000年2月，美国参议院的一个委员会认定，这些支出是合理的。

许多人却持不同看法，认为威胁被夸大了，甚至是个骗局。但局面之所以平稳，很大程度上正是因为大量问题已经修好。这就是“防备悖论”：预防措施发挥作用，使所抵御的危险没有出现，危险反而看起来像是凭空捏造的。繁琐的修复工作换来的不是感谢，而是嘲笑。

批评者指出，意大利和韩国花的钱少得多，结果却并不更差。但这种比较具有误导性：较低的支出数字往往没有算上企业的花费，而且很多修补工作也替换了早该升级的过时系统。没人能逐笔说清，哪些开支必不可少，哪些花得过多。奏效的预防可能只留下账单，人们只能拿它与那场本可能发生的灾难作比较。

文中的约1000亿美元是当时的估计；平稳过渡不代表每笔支出都已被证明必要。

核心词汇

先记这22项。连着下方的原文短语一起回想。

bug n.
程序漏洞
The Y2K bug

shortcut n.
捷径；省事的办法
began as a shortcut

ambiguity n.
歧义
That compact code hid an ambiguity

corrupt v.
使出错；破坏
corrupting calculations

brace for 动词短语
准备应对（困难）
firms braced for chaos

coordinate v.
协调
to coordinate America's effort

audit v.
审查；核查
Consultants audited legacy systems

legacy adj.
沿用的；旧有的
legacy systems

patch v.
修补（程序）
patched faulty date logic

projection n.
预测
projections of the global bill

sporadic adj.
零星的；偶发的
sporadic and mostly minor

expenditure n.
支出
the expenditure had been justified

justified adj.
合理的；有正当理由的
the expenditure had been justified

exaggerate v.
夸大
the threat had been exaggerated

hoax n.
骗局
even a hoax

preparedness paradox n.
防备悖论
the preparedness paradox

precaution n.
预防措施
when precautions work

ward off 动词短语
抵御
the danger they ward off

omit v.
遗漏
omitted what firms spent

obsolete adj.
过时的
replaced obsolete systems

essential adj.
必不可少的
which dollars were essential

catastrophe n.
灾难
a catastrophe that might have been

文中词汇与搭配

补充词汇共41项。保留每个表达在本文中的意思。

millennium n.
千年；千禧年

memory n.
计算机内存

hardware n.
硬件

costly adj.
昂贵的

programmer n.
程序员

digit n.
数字

compact adj.
精简紧凑的

code n.
程序代码

deadline n.
截止日期

chaos n.
混乱

president n.
总统

council n.
委员会

consultant n.
顾问

faulty adj.
有缺陷的

Commerce Department n.
美国商务部

enterprise n.
企业

global adj.
全球的

midnight n.
午夜

strike v.
（钟）敲响

glitch n.
小故障

文中词汇与搭配 续

补充词汇共41项。保留每个表达在本文中的意思。

crop up 动词短语
意外出现

minor adj.
轻微的

falter v.
失灵；运转不稳

spy n.
间谍

conclude v.
得出结论

prevail v.
占上风；普遍存在

largely adv.
很大程度上

imaginary adj.
想象出来的

tedious adj.
繁琐乏味的

mockery n.
嘲笑

gratitude n.
感激

South Korea n.
韩国

fare no worse 动词短语
情况并不更差

comparison n.
比较

mislead v.
误导

remedial adj.
补救的

overdue for 形容词短语
早该进行……的

excessive adj.
过多的；过度的

prevention n.
预防

invoice n.
账单

be weighed against 动词短语
与……权衡比较

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That compact code hid an _____: a machine might read "00" as 1900, corrupting calculations of interest, ages, or deadlines.

02 Consultants audited legacy systems and _____ faulty date logic.

03 Glitches that did crop up were _____ and mostly minor.

04 This is the preparedness paradox: when _____ work, the danger they ward off never appears, so it looks imaginary.

文章主线

- 隐患 旧程序为省内存只用两位数字记年，可能将2000年的“00”误读为1900年，算错利息、年龄和期限。
- 应对 政府与企业排查旧系统、修补日期逻辑；美国1999年预计花约1,000亿美元，全球预测达数千亿美元。
- 跨年 2000年跨年大体平稳，故障零星轻微；美国参议院委员会认定开支合理，却有人把威胁当成骗局。
- 悖论 平稳很大程度上来自预先修复：预防奏效、灾难未现，危险反倒像凭空捏造，这就是“防备悖论”。
- 限度 意大利、韩国的低支出统计常漏算企业花费，修补也更新了过时系统；究竟哪些钱必要、哪些过量，无法逐笔断定。

参考答案　1 ambiguity 2 patched 3 sporadic 4 precautions